

Metric lengths change the Jacobian lattice determinant

Disproof of Conjecture 00000008356

gaochengzhecpu

Abstract

A triangle whose three edges have length $1/2$ has Jacobian cycle-lattice Gram determinant $3/2$, whereas its ordinary spanning-tree count is 3. Thus the stated unweighted equality fails for metric graphs without a unit-length normalization. The alternative convention of lattice determinant as Euclidean covolume gives $\sqrt{3/2}$ and also fails.

Definitions and scope

The source asserts that the determinant of the Jacobian lattice of a metric graph equals its spanning-tree count. Here the latter has its ordinary meaning: each spanning tree of the specified finite graph is counted once. The source imposes no unit-length normalization.

For an oriented metric graph with positive edge lengths ℓ_e , the integral cycle lattice is

$$L = H_1(G, \mathbb{Z}) = \ker(\partial : \mathbb{Z}^E \longrightarrow \mathbb{Z}^V), \quad Q(z, w) = \sum_{e \in E} \ell_e z_e w_e.$$

The real cycle space modulo this lattice, with this length pairing, is the metric Jacobian model. For an integral basis $b_1, \dots, b_g$, the Gram determinant is $\det(Q(b_i, b_j))$. It is independent of the integral basis because a basis change has determinant ± 1 . This is the usual metric length pairing; see Mikhalkin–Zharkov, Section 6.1 and Lemma 6.1¹.

The counterexample

Let $G = K_3$ on vertices $0, 1, 2$, with oriented edges $e_0 : 0 \rightarrow 1$, $e_1 : 1 \rightarrow 2$, and $e_2 : 2 \rightarrow 0$. Set every $\ell_e = 1/2$. In these coordinates,

$$\partial(z_0, z_1, z_2) = (z_2 - z_0, z_0 - z_1, z_1 - z_2).$$

Consequently $\partial z = 0$ exactly when $z_0 = z_1 = z_2$, so $L = \mathbb{Z}(1, 1, 1)$ and $b = (1, 1, 1)$ is an integral basis. Its length Gram matrix and determinant are

$$[Q(b, b)] = [\tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2}] = [\tfrac{3}{2}], \quad \det_j(G) = \tfrac{3}{2}.$$

Every spanning tree of the triangle has exactly two edges, and every two-edge subset is a spanning tree. There are exactly three such subsets. Hence

$$\boxed{\det_j(G) = \tfrac{3}{2} \neq 3 = \tau(G).}$$

If determinant denotes Euclidean covolume instead, the left side is $\sqrt{3/2} \neq 3$, since equality would imply $3/2 = 9$.

¹G. Mikhalkin and I. Zharkov, *Tropical curves, their Jacobians and Theta functions*.

The missing lengths are essential. For this triangle the metric tree-complement expression is $\sum_T \prod_{e \notin T} \ell_e = 3/2$; it counts each omitted edge with its length and is not the unweighted tree count 3. With unit edge lengths the Gram determinant would indeed be 3. Thus the counterexample concerns exactly the unnormalized metric-graph equality stated in the source. Replacing its right side by a weighted enumerator gives a different statement. Failure of the first asserted equality suffices; the other clauses are not needed.

Correspondence with Lean

The Lean object `K3` is an actual `SimpleGraph`. The subtype `SpanningTree` consists of its spanning connected acyclic subgraphs, and `spanning_tree_count` proves its cardinality is 3 via an exhaustive finite edge-mask bijection. The theorem `oriented_edges_are_actual` checks that the chosen oriented edges enumerate the edges of that same graph.

The linear map `boundary` comes from the actual incidence matrix. Its kernel `CycleLattice` is equipped with an explicit linear equivalence to $\mathbb{Z}$, giving the actual Mathlib `Basis` named `cycleBasis`. The matrix `gram` evaluates the edge-length pairing on this basis. The final theorems `determinant_ne_spanning_tree_count` and `covolume_ne_spanning_tree_count` establish both convention-specific inequalities. The formalized object is the metric cycle lattice; no unrelated scalar is assigned the name “Jacobian determinant.”

Reproduction

The project pins Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build` and `lake env lean -DwarningAsError=true Main.lean`. On a new machine, `lake exe cache get` retrieves the official pinned dependency artifacts. The submission itself is checked in a fresh build directory. Printed dependencies use only the standard `propext`, `Classical.choice`, and `Quot.sound`. No additional axioms, proof placeholders, or numerical oracles are used. Run `tectonic main.tex` to reproduce the PDF; no auxiliary numerical script is needed. Build logs, source hashes, and visual-review records are included under `verification/`.

Conjecture source

The Last Math Competition, Conjecture 00000008356. The exact bilingual source and its commit and SHA-256 are preserved in this submission.